Algebra II with Trigonometry Chapter 6.4

Mental Math (Gemini)

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Basic Periods 1. What is the fundamental period of the parent tangent and cotangent functions?

Secondary Periods 2. Which two trigonometric functions discussed in the text have a fundamental period of 2π?

Shared Asymptotes 3. Which two functions share vertical asymptotes at $x = \frac{\pi}{2} + n\pi$ (where n is an integer)?

Cosecant Asymptotes 4. The cotangent function has vertical asymptotes at $x = n\pi$. Which other parent function shares these exact same vertical asymptotes?

Reciprocal Identities 5. The graph of the cosecant function uses the identity $\csc x = \frac{1}{\sin x}$. What identity is used to graph the secant function?

Appropriate Intervals 6. According to the text, what is the appropriate interval used to graph exactly one period of $y = a \cot kx$ (assuming no phase shift)?

Transformations I (Tangent) 7. Looking at Example 1a, what is the period of the function $y = 4\tan(4x - 2\pi)$?

Transformations II (Cosecant) 8. Looking at Example 1b, what is the period of the function $y = 2\csc(\pi x - \frac{\pi}{3})$?

Transformations III (Secant) 9. Looking at Example 1c, what is the period of the function $y = 5\sec(3x - \frac{\pi}{2})$?

Transformations IV (Cotangent) 10. Looking at Example 1d, what is the period of the function $y = 2\cot(3\pi x + 3\pi)$?